

RESEARCH INTERESTS

Mechanistic Interpretability of LLMs, LLM Alignment, Cultural and Low Resource NLP, AI Safety

EDUCATION

Jahangirnagar University

Master of Science in Computer Science (PMSCS track)

Dhaka, Bangladesh

May 2026 (Expected)

▪ **CGPA:** 3.97/4.0

▪ **Relevant Coursework:**

- * **Machine Learning & AI:** Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Artificial Intelligence, Computational Intelligence
- * **Core Computer Science:** Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Computer Architecture
- * **Systems & Software:** Programming Languages, Software Engineering

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Civil Engineering (B.Sc Engg.)

Dhaka, Bangladesh

Mar. 2025

▪ **CGPA:** 3.77/4.0, **Major:** Transportation Engineering, **Minor:** Environmental Engineering

▪ **Relevant Coursework:**

- * **Mathematics:** Differential and Integral Calculus, Matrices, Laplace Transform and Vector Analysis, Numerical Methods, Applied Mathematics for Engineers
- * **Statistics:** Probability & Statistics
- * **Programming:** Computer Programming Sessional (Structured Programming, OOP), Engineering Computation
- * **Applied:** Transportation Engineering, Structural Analysis, Urban Transportation Planning

RESEARCH EXPERIENCE

Graduate Researcher

JU Data Mining Research Group

August 2025 – Present

Supervisor: Dr. Md Musfique Anwar

o **Event Detection in Bangla Text: Assessing Language Model Robustness to Noisy Data**

Research Project, Manuscript in preparation

- * Developed a robust event detection framework for Bangla to benchmark the performance of instruction-tuned LLMs against encoder-based models on noisy, ASR-transcribed news text.
- * Synthesized perturbed datasets to quantify model degradation thresholds, specifically analyzing the impact of Word Error Rate (WER) on the stability of event trigger extraction.

o **BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction**

Jan – May

Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar

Accepted (Oral) at ACL Student Research Workshop (SRW) 2026 [\[arXiv\]](#)

- * Conceptualized and led the development of BanglaSocialBench, a benchmark comprising 1,719 culturally grounded instances across three domains to evaluate LLM sociopragmatic competence.
- * Benchmarked 12 LLMs in zero-shot setting, uncovering over-politeness, religious kinship term conflation, and rigidity in ambiguous social contexts.

Undergraduate Thesis

Bangladesh University of Engineering and Technology (BUET)

Nov. 2023 – Feb. 2025

Supervisor: Dr. Md. Shamsul Hoque

o **Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data**

Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque

Submitted to Accident Analysis & Prevention, 2025

- * Built an end-to-end multimodal data pipeline integrating Google Street View API, Overture Maps, OpenStreetMap, and crash datasets to generate urban risk features for pedestrian crash severity prediction.
- * Applied semantic segmentation (Detectron2 + Mask2Former) on 12K+ street-view images and trained ensemble ML models (XGBoost, Random Forest, AdaBoost, CatBoost) with SHAP-based interpretability to identify key environmental risk factors.

WORK EXPERIENCE

Maks Inc.

Research Intern

Dhaka, Bangladesh (HQ: Frisco, TX, USA)

May 2025 – Sep 2025

Projects: World Bank Heavy-Duty Vehicles (HDV) Baseline Assessment, Akhaura–Tamabil Land Port Feasibility Study

- Modeled baseline HDV air pollution metrics using the MOVES5 and large-scale BRTA fleet data.
- Used QGIS for spatial analysis and supported field-data collection for infrastructure assessment studies.
- Conducted stakeholder interviews across 7 administrative divisions with transportation operators, associations, and local leaders to document operational and socioeconomic conditions.
- Performed literature reviews and contributed to technical reports with project engineers and research scientists.

HONORS & AWARDS

Undergraduate Student Research Grant

Jan. 2024

Research and Innovation Centre for Science and Engineering, BUET

- Awarded competitive research funding for undergraduate research project on **Deep Learning-based analysis and SHAP-based interpretability** of pedestrian crash severity using public geospatial data.

Dean's List Scholarship

Jul. 2019

Bangladesh University of Engineering and Technology (BUET)

- Awarded for maintaining an average GPA of 3.75 or above across two consecutive regular terms.

3rd Place, Concrete Solutions Competition

Mar. 2022

American Concrete Institute (ACI)

- Recognized for the most innovative team design; project featured in the **Daily Prothom Alo** [\[Feature\]](#)

Higher Secondary Certificate (HSC) Scholarship

2019

Board of Intermediate and Secondary Education, Rajshahi

- Awarded **Talentpool Scholarship**; ranked **10th (top 0.01%)** among all candidates in the Rajshahi Board.

ACADEMIC PROJECTS

Mohr's Circle and 2D Stress Calculator App using MATLAB

Engineering Computation Sessional

- Developed an interactive application using MATLAB App Designer that can plot Mohr's Circle from stress block inputs, identify max/min stresses, display plane orientations, and export plots. [\[GitHub\]](#)

LEADERSHIP & VOLUNTARY WORKS

ITE BUET Student Chapter

Dec. 2024 – Mar. 2025

Deputy Vice President

- Collaborated with other executive members to organize workshops and seminars on various topics of transportation research.

ACI BUET Student Chapter

Mar. 2024 – Mar. 2025

Vice President

- Organized poster competitions, seminars, and field visits in coordination with chapter members.

SKILLS

Technical Skills

- **Languages:** C, C++, Python, JavaScript, SQL
- **Libraries:** NumPy, PyTorch, Matplotlib, scikit-learn, Selenium
- **Other:** Git, LaTeX, Bash